

MegaSquirt + ArduPilot

EFI integration via CAN Bus

Installation and configuration guide

Script Lua: EFI_Megasquirt_CAN.lua | ArduPilot 4.3+

1. Introduction

This document describes how to install and configure the Lua driver `EFI_Megasquirt_CAN.lua` on an ArduPilot autopilot (Pixhawk / Cube / similar) to receive telemetry from the MegaSquirt 2 / MicroSquirt / MS2Extra ECU in real time via the CAN bus.

The script receives the engine data and publishes it to the ArduPilot EFI backend, making variables such as RPM, intake pressure, air temperature, coolant temperature, battery voltage, and AFR available as MAVLink telemetry and in the flight logs.

NOTE: This script is for receiving data only. It does not send commands to the ECU or modify any MegaSquirt parameters.

2. Requirements

2.1 Hardware

- **Autopilot:** Pixhawk 4 / Cube Orange / any ArduPilot board with a physical CAN port
- **ECU:** Loweheiser ECU1 or ECU4
- **Bus CAN:** 2-wire cable (CAN_H and CAN_L) + common GND between ECU and autopilot
- **Termination:** Exactly two 120-ohm resistors (one at each end of the bus)
- **microSD card:** Installed on the autopilot to store the script

IMPORTANT: Tanto the Loweheiser ECU as The Pixhawk or other controllers usually have a 120ohm termination resistor and therefore nothing needs to be added; if communication fails then, vVerify that the total resistance between CAN_H and CAN_L of the bus (with everything connected and turned off) is ~60 ohms.

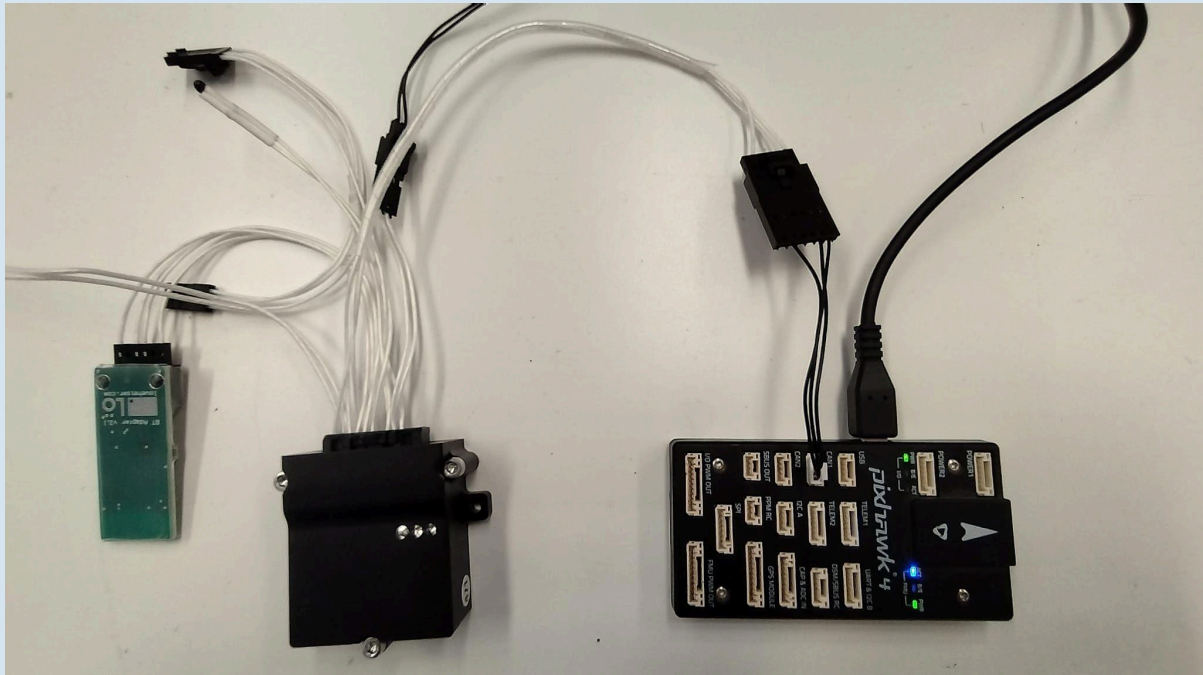


PHOTO 1: CAN connection diagram — ECU <-> Pixhawk (CAN_H, CAN_L, GND wires and resistors at each end)

2.2 Software

- **ArduPilot:** Version 4.3 or superior (ArduCopter, ArduPlane, ArduRover, etc.)
- **Mission Planner:** Version 1.3.80 or higher (to configure parameters and view telemetry)
- **TunerStudio:** With the ECU connected to configure the CAN broadcast
- **File:**EFI_Megasquirt_CAN.lua (included in this package)

3. MegaSquirt Configuration (TunerStudio)

Perform these steps with TunerStudio connected to the ECU.

3.1 Enable CAN Realtime Data Broadcasting

1. Open TunerStudio and connect to the ECU.
2. In the menu **CAN Bus / Testmodes** and activate real-time broadcasting


CAN Parameters
×

File View Help

CAN Parameters


My CAN ID


Master Enable


Enable PWM Polling


Remote CAN ID


Remote Table Number For PWM data


Remote Table Offset For PWM data(bytes)


Remote PWM Clock Frequency(MHz)


Remote PWM Clock Prescale


Enable Ports Polling


Remote CAN ID


Remote Table Number For Ports Data


Remote Table Offset For Ports Data(bytes)


Remote Ports Direction


Remote Port 1 Type


Remote Port 2 Type


Remote Port 3 Type


Use For Remote On/Off Outputs

CAN ADC Selection

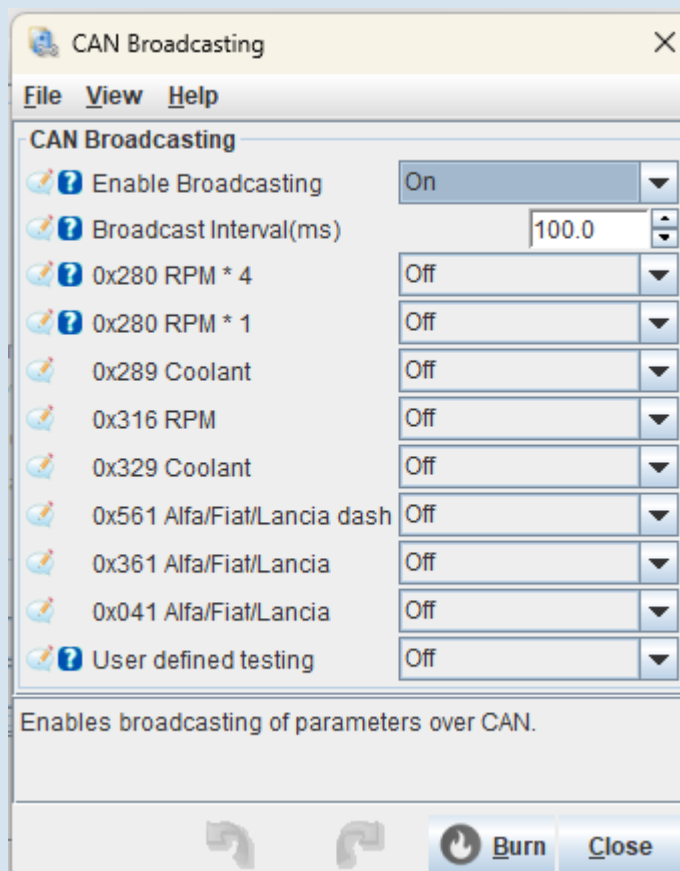
CAN ADC Group On/Off	CAN Id	Table	Offset
 CAN ADC 0-3 Disable	 5	 7	 (bytes) 2
 CAN ADC 4-7 Disable	 5	 7	 (bytes) 10

Set to 0 unless specifically wanting to configure this as a secondary data capture ECU.




Burn

Close



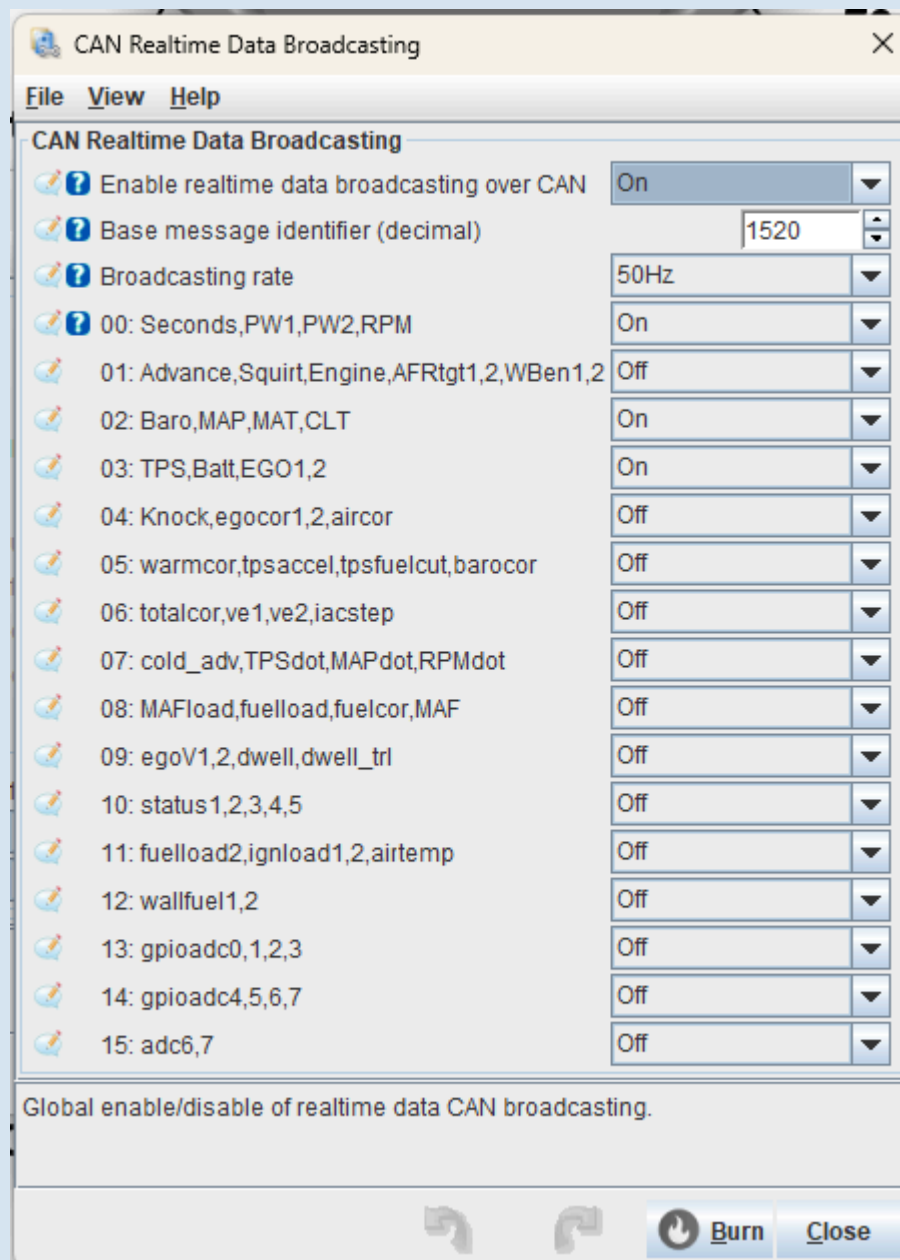


PHOTO 2: Screenshot from TunerStudio —Compatible with Loweheiser ECU1

4. Installing the script in the autopilot

4.1 Copy the file to the SD card

The script must be in a specific location within the autopilot's microSD card:

```
APM/
scripts/
  EFI_Megasquirt_CAN.lua
```

3. Remove the SD card from the autopilot and insert it into the PC (or use MAVFtp from Mission Planner — see step 4.2).
4. Create the folder **APM/scripts/** if it doesn't exist.
5. Copy the file **EFI_Megasquirt_CAN.lua** inside that folder.
6. Safely remove the SD card and reinsert it into the autopilot.

The path is case-sensitive. It must be **APM/scripts/** (not **Scripts/**, not **scripts** in the root directory). An incorrect path will prevent the script from loading without displaying any visible error message.

4.2 Alternative: copy via MAVFtp (without removing the SD card)

7. As a Mission Planner, pulse **Ctrl+F** and search **MAVFtp**.
8. Navigate to **APM/scripts/** in the tree on the left.
9. Upload the file **EFI_Megasquirt_CAN.lua** with the Upload button.

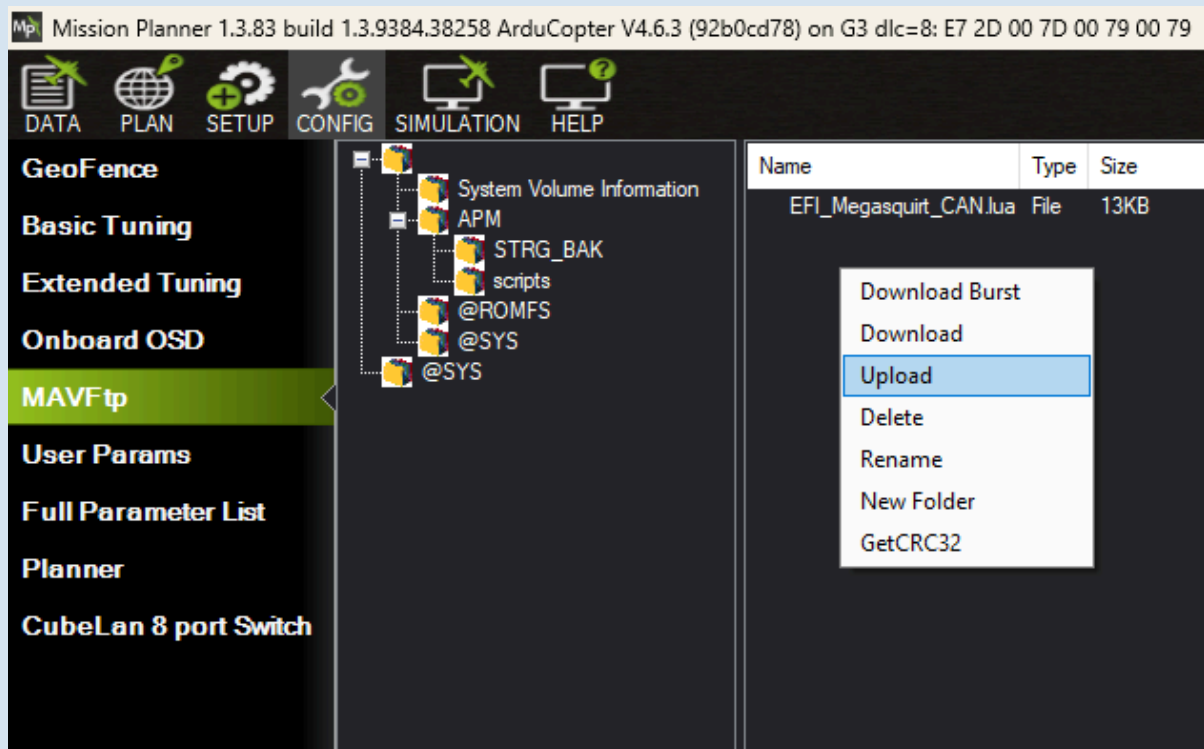


FOTO 4: Captura de Mission Planner — MAVFtp mostrando APM/scripts/EFI_Megasquirt_CAN.lua con 11KB de tamaño

5. Configuring parameters in ArduPilot

Go to Mission Planner → Config → Full Parameter List, find each parameter and set the indicated values. After changing all the parameters, press Write Params and then perform a physical reboot (disconnect and reconnect the power).

5.1 General system parameters

Parameter	Value	Description
SCR_ENABLE	1	Activate the Lua scripting engine
SCR_HEAP_SIZE	90000	Memory reserved for scripts (bytes). Do not exceed 100,000 in Pixhawk4
EFI_TYPE	7	Select the EFI backend by Lua Scripting

5.2 CAN bus parameters

Parameter	Value	Description
CAN_P1_DRIVER	1	Assign the physical CAN1 port to driver 1
CAN_D1_PROTOCOL	10	Scripting protocol (necessary for Lua to access CAN)
CAN_P1_BITRATE	500000	CAN speed — must match TunerStudio (500 kbit/s)

If the physical cable goes to the autopilot's CAN2 connector, use CAN_P2_DRIVER and CAN_D2_PROTOCOL instead of the _P1_ and _D1_ values indicated above, and change EFI_MS_CANDRV to 2.

5.3 EFI MegaSquirt Driver Parameters

These parameters appear automatically in the list **ONLY afterwards** from the first boot with the script on the SD and SCR_ENABLE=1. If they do not appear, check steps 4 and 5.1 first.

Parameter	Value	Description
EFI_MS_ENABLE	1	Activate the driver. Without this, the script will silently deactivate.
EFI_MS_CANDRV	1	CAN driver to use (1 = CAN1, 2 = CAN2)
EFI_MS_MODE	1	Decoding mode: 1 = Advanced Realtime Broadcast (the one we use)
EFI_MS_BASE_ID	1520	CAN base ID decimal — must match TunerStudio
EFI_MS_RATE_HZ	50	Script update frequency (Hz)
EFI_MS_NEW_KPA	101.3	Reference atmospheric pressure (for mode 0 only)

EFI_MS_OPTIONS

0

Additional options. Set to 1 to enable CAN
frame logging in .bin

6. Verification of operation

6.1 Confirm that the script has loaded

10. Connect Mission Planner to the autopilot with the ECU powered on and the SD card inserted.
11. Open the tab **Messages** (sandwich icon or HUD menu).
12. Perform a physical reboot of the autopilot (disconnect/connect power).
13. In the first few seconds of startup, it looks for the line:

```
EFI: MegaSquirt CAN loaded
```

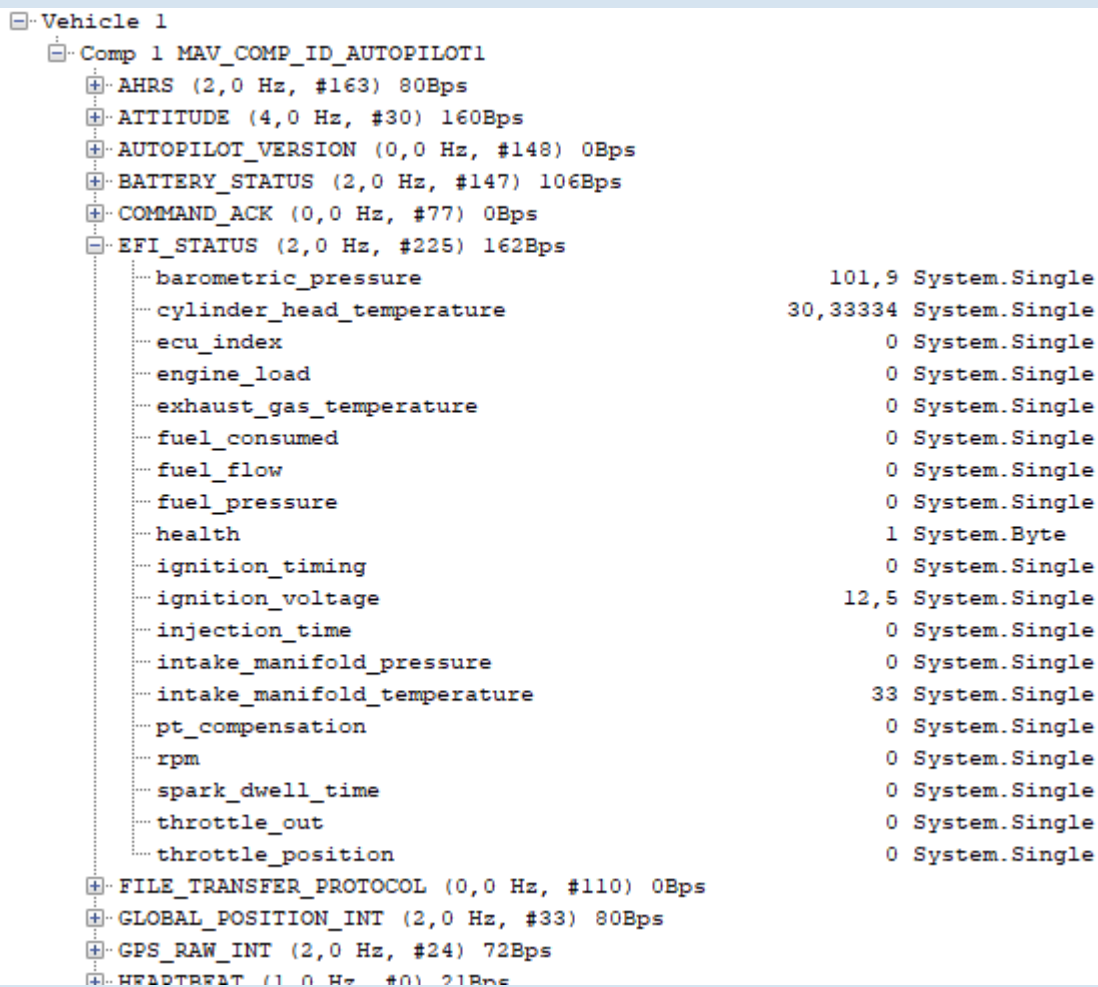
If this line appears, the script has loaded correctly.

```
07/07/2026 11:57:52 : G3 dlc=8: E9 6F 00 7E 00 79 00 79
07/07/2026 11:57:52 : G2 dlc=8: 03 FB 00 00 03 8F 03 85
07/07/2026 11:57:52 : G0 dlc=8: 02 6D 00 00 00 00 00 00
07/07/2026 11:57:52 : .6V tps=-577.7
07/07/2026 11:57:52 : EFI dbg: rpm=0 CLT=32.3C MAT=32.8C MAP=0.0 batt=12
07/07/2026 11:57:52 : EFI: MegaSquirt CAN: frames=747 backend=true
07/07/2026 11:57:47 : Frame: HEXA/X
07/07/2026 11:57:47 : IMU0: fast sampling enabled 8.0kHz/2.0kHz
07/07/2026 11:57:47 : RCOut: PWM:1-12 PWM:15-16
07/07/2026 11:57:47 : IOMCU: 420 1001 411FC231
07/07/2026 11:57:47 : Pixhawk4 002D0035 34385119 39313933
07/07/2026 11:57:47 : ChibiOS: 88b84600
07/07/2026 11:57:47 : ArduCopter V4.6.3 (92b0cd78)
07/07/2026 11:57:47 : RAW: id=0x5F3 mode=1 base=1520 dlc=8
07/07/2026 11:57:47 : RAW: id=0x5F2 mode=1 base=1520 dlc=8
07/07/2026 11:57:47 : RAW: id=0x5F0 mode=1 base=1520 dlc=8
07/07/2026 11:57:47 : tps=0.0
07/07/2026 11:57:47 : EFI dbg: rpm=0 CLT=0.0C MAT=0.0C MAP=0.0 batt=0.0V
07/07/2026 11:57:47 : EFI: MegaSquirt CAN: frames=0 backend=true
07/07/2026 11:57:47 : EFI: MegaSquirt CAN: EFI backend OK
07/07/2026 11:57:47 : EFI: MegaSquirt CAN loaded
07/07/2026 11:57:46 : Frame: HEXA/X
07/07/2026 11:57:46 : IMU0: fast sampling enabled 8.0kHz/2.0kHz
07/07/2026 11:57:46 : RCOut: PWM:1-12 PWM:15-16
07/07/2026 11:57:46 : IOMCU: 420 1001 411FC231
07/07/2026 11:57:46 : Pixhawk4 002D0035 34385119 39313933
07/07/2026 11:57:46 : ChibiOS: 88b84600
07/07/2026 11:57:46 : ArduCopter V4.6.3 (92b0cd78)
07/07/2026 11:57:46 : Frame: HEXA/X
07/07/2026 11:57:46 : IMU0: fast sampling enabled 8.0kHz/2.0kHz
07/07/2026 11:57:46 : RCOut: PWM:1-12 PWM:15-16
07/07/2026 11:57:46 : IOMCU: 420 1001 411FC231
07/07/2026 11:57:46 : Pixhawk4 002D0035 34385119 39313933
07/07/2026 11:57:46 : ChibiOS: 88b84600
07/07/2026 11:57:46 : ArduCopter V4.6.3 (92b0cd78)
07/07/2026 11:57:46 : RCOut: PWM:1-12 PWM:15-16
07/07/2026 11:57:46 : ArduPilot Ready
07/07/2026 11:57:46 : Frame: HEXA/X
07/07/2026 11:57:46 : IMU0: fast sampling enabled 8.0kHz/2.0kHz
```


PHOTO 5: Screenshot of Mission Planner — Messages tab showing 'EFI: MegaSquirt CAN loaded' after boot and other information that the message gives as raw data and decoded data

6.2 Verify data in real time

14. Open the **Mavlink Inspector** (Ctrl+F → 'Mavlink Inspector').
15. Expand the message **EFI_STATUS** (#225).
16. With the ECU switched on and the engine running, check that the following fields display real values:



```
Vehicle 1
  Comp 1 MAV_COMP_ID_AUTOPILOT1
    AHRS (2,0 Hz, #163) 80Bps
    ATTITUDE (4,0 Hz, #30) 160Bps
    AUTOPILOT_VERSION (0,0 Hz, #148) 0Bps
    BATTERY_STATUS (2,0 Hz, #147) 106Bps
    COMMAND_ACK (0,0 Hz, #77) 0Bps
    EFI_STATUS (2,0 Hz, #225) 162Bps
      barometric_pressure 101,9 System.Single
      cylinder_head_temperature 30,33334 System.Single
      ecu_index 0 System.Single
      engine_load 0 System.Single
      exhaust_gas_temperature 0 System.Single
      fuel_consumed 0 System.Single
      fuel_flow 0 System.Single
      fuel_pressure 0 System.Single
      health 1 System.Byte
      ignition_timing 0 System.Single
      ignition_voltage 12,5 System.Single
      injection_time 0 System.Single
      intake_manifold_pressure 0 System.Single
      intake_manifold_temperature 33 System.Single
      pt_compensation 0 System.Single
      rpm 0 System.Single
      spark_dwell_time 0 System.Single
      throttle_out 0 System.Single
      throttle_position 0 System.Single
    FILE_TRANSFER_PROTOCOL (0,0 Hz, #110) 0Bps
    GLOBAL_POSITION_INT (2,0 Hz, #33) 80Bps
    GPS_RAW_INT (2,0 Hz, #24) 72Bps
    HEARTBEAT (1,0 Hz, #0) 21Bps
```

PHOTO 6: Screenshot from Mission Planner — Mavlink Inspector showing expanded **EFI_STATUS** with actual RPM, temperature, and pressure values

7. Problem Solving

7.1 'EFI: MegaSquirt CAN loaded' does not appear on startup

- **SCR_ENABLE = 0:** Check the parameter and set it to 1. Reboot required.
- **File not found:** Verify via MAVFtp that the file is in APM/scripts/ with the exact name.
- **Out of memory.** Reduce SCR_HEAP_SIZE to 90000 and ensure there are no other scripts on the SD card. Check MEMINFO in Mavlink Inspector — freemem32 must be greater than SCR_HEAP_SIZE.
- **EFI_MS_ENABLE = 0:** The default parameter is 0. Set it to 1 after the first boot where the EFI_MS_* parameters appear.

7.2 The script loads but EFI_STATUS shows everything as 0

- **EFI_TYPE != 7:** The EFI backend does not initialize without EFI_TYPE=7.
- **Incorrect EFI_MS_BASE_ID:** It must be 1520, not 1512. Confirm in TunerStudio that the Base message identifier is exactly 1520.
- **EFI_MS_MODE = 0:** It must be 1 for Advanced Realtime Broadcast. With 0, expect a different format.
- **CAN bus with no activity:** Connect a CAN analyzer (CANgaroo + USB-CAN adapter) to the bus and check that frames with IDs 0x5F0, 0x5F2 and 0x5F3 are arriving.

7.4 The CAN bus is not working

- **Incorrect ending:** Measure with a multimeter between CAN_H and CAN_L (power off). It should read approximately 60 ohms. If it reads 120, a resistor is missing. **CAN_H and CAN_L crossed:** Try swapping the two wires between the ECU and the autopilot.
- **Common table:** The CAN bus requires a shared ground (GND) between the ECU and the autopilot. Connect a ground wire directly between them if you don't already have one.
- **Incorrect bitrate:** Both ends should be at 500000 bps. Check in TunerStudio and in CAN_P1_BITRATE.